Hotel Booking Data Analysis and Business Intelligence Dashboard

Occupancy, cancellations and revenue for two hotels, built in Power BI

Presented by Your Name

34,407
bookings analysed

₹38.29 Cr
confirmed revenue

30.8%
of bookings cancelled

Why this project: cancellations cost revenue

30.8%
of bookings are cancelled

₹21.28 Cr
revenue lost to cancellations

52.6%
average occupancy

The problem

Hotels earn money only when rooms are filled at a good rate, yet about one booking in three is cancelled. Management has raw booking records but no single view of occupancy, cancellations and revenue.

Objectives

- Clean and model the booking data as a star schema
- Measure bookings, revenue, ADR, occupancy and RevPAR
- Find which guests, channels and lead times drive cancellations
- Deliver an interactive dashboard with clear recommendations

The dataset: two hotels over three years

2
hotels

2022 to 2024
arrival period

31
columns per booking

INR
currency

Simulated, realistic data built for this project, with planted errors to clean.

File	Rows	Description
hotel_bookings.csv	34,662	Fact table, one row per booking
dim_hotel.csv	2	Hotel name, type, location, rooms
dim_room_type.csv	4	Room type code and description
dim_country.csv	10	Country code, name and region

Hotels: Chennai City Hotel (60 rooms) and ECR Beach Resort (40 rooms).

Cleaning removed 255 faulty rows

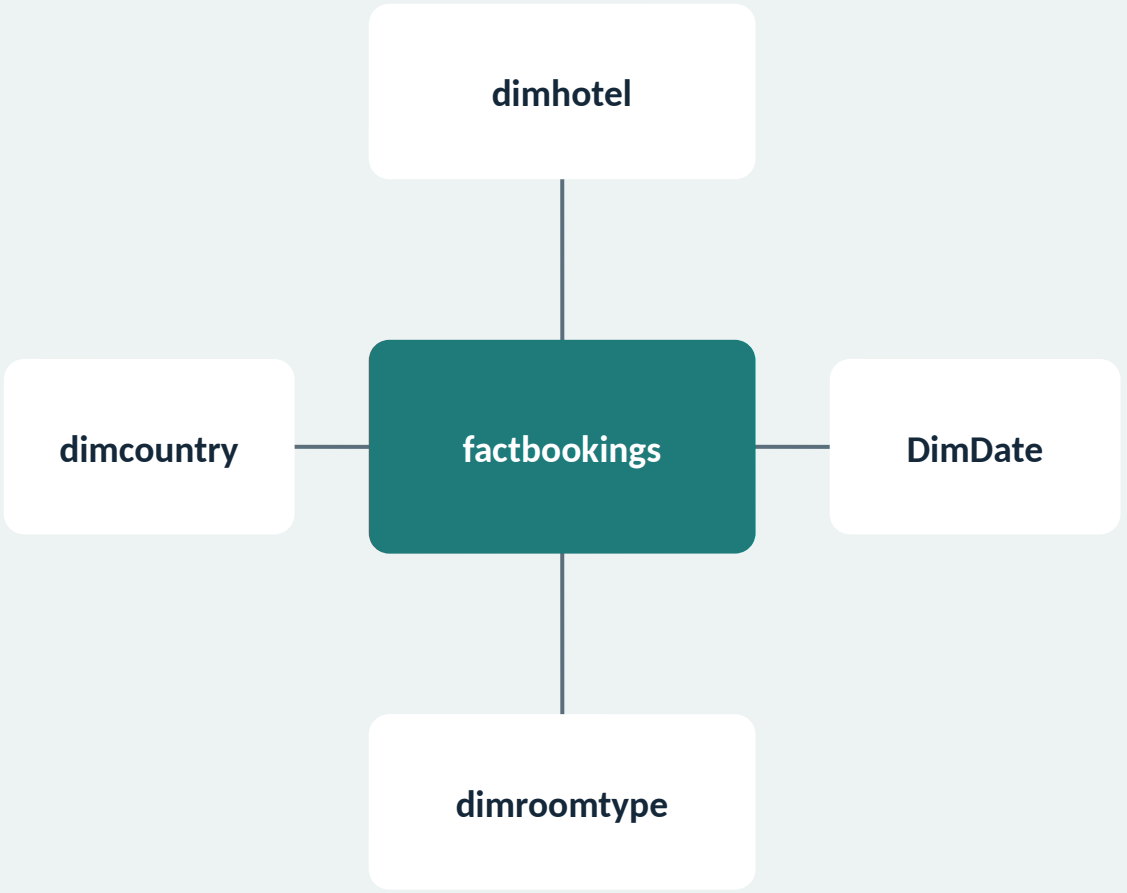
34,662
raw rows

34,407
clean rows

Cleaning step	Rows
Removed duplicate bookings	-150
Removed zero-guest bookings	-60
Removed negative prices	-25
Removed extreme price outliers	-20
Blank children replaced with 0	300
Blank country set to Unknown	about 500
Added 6 columns, such as Total Guests and Booking Revenue	new

Zero-price stays were kept: they are valid complimentary bookings.

Star schema and DAX measures



One fact table, four dimension tables

Cancellation Rate
Cancelled bookings ÷ total bookings

Total Revenue
ADR × nights, confirmed bookings only

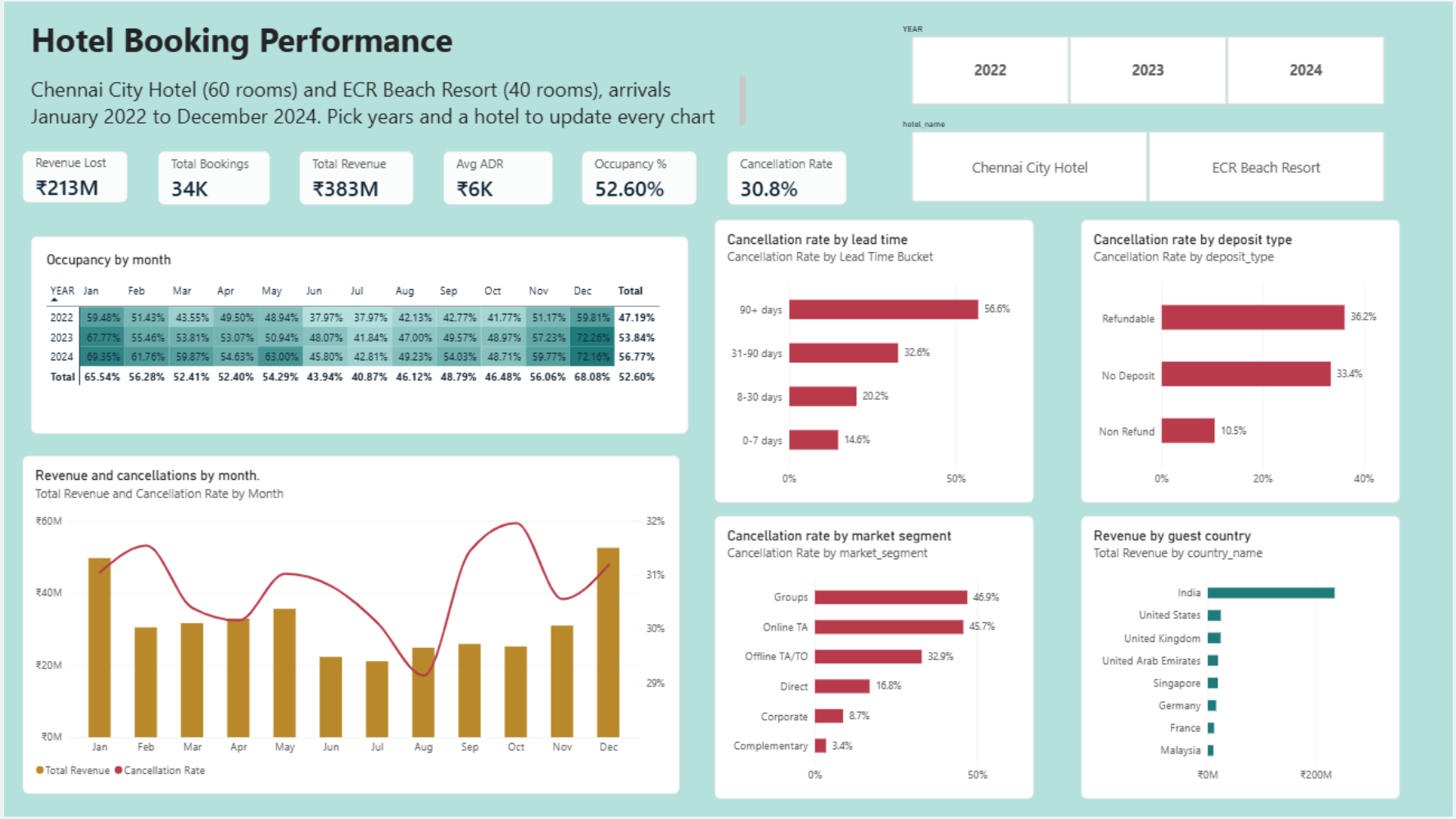
Avg ADR
Average daily rate of confirmed stays

Occupancy %
Room-nights sold ÷ rooms available

RevPAR
Revenue ÷ rooms available

Revenue Lost
ADR × nights, cancelled bookings

The Power BI dashboard



Slicers

Year and hotel filter every card and chart

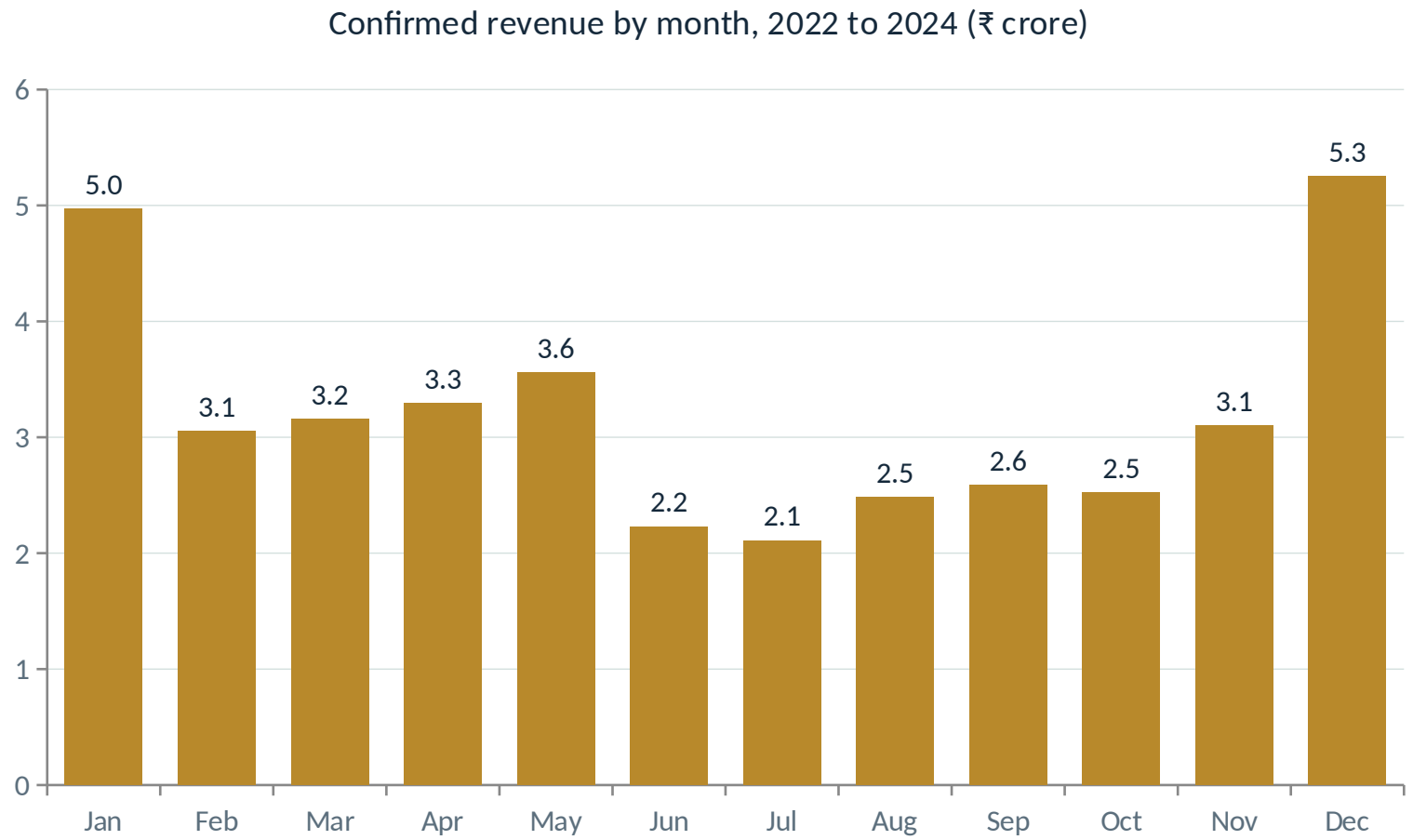
KPI cards

Revenue, bookings, ADR, occupancy and cancellation rate

Charts

Occupancy heatmap, monthly trend, cancellations, countries

Seasonality: December peaks, July dips



December: ₹5.25 Cr

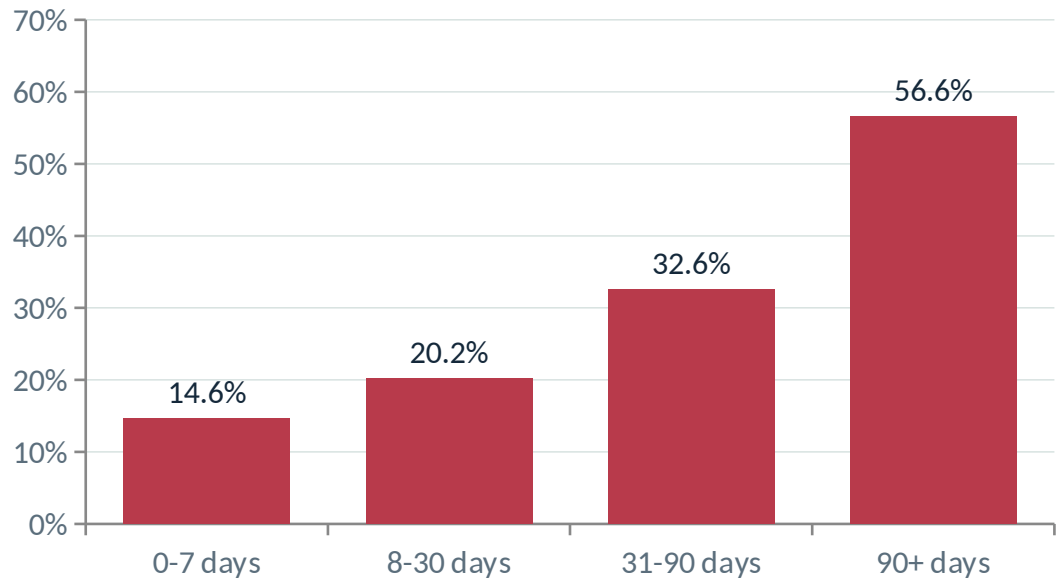
strongest month

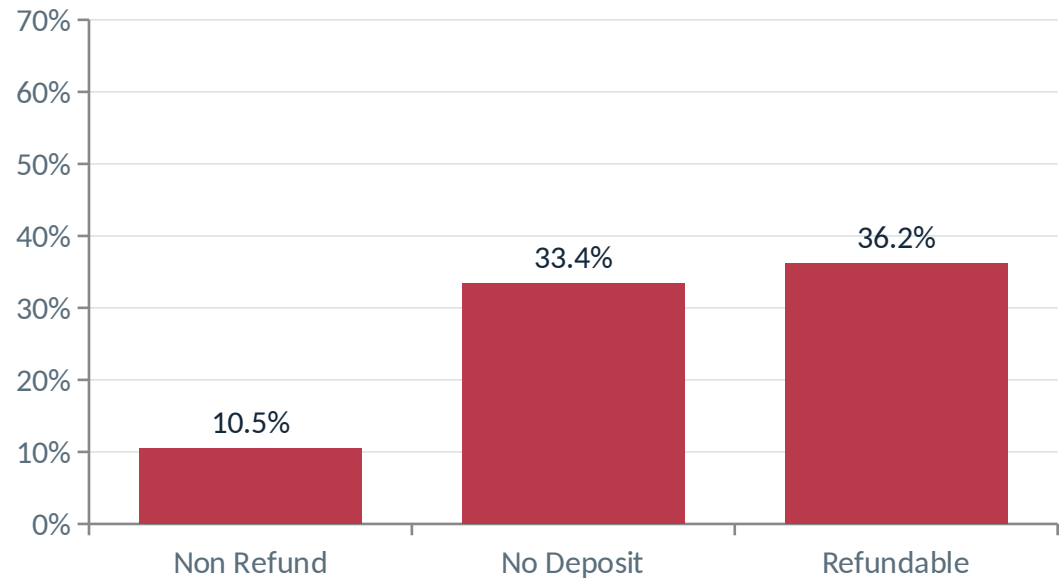
July: ₹2.11 Cr

weakest month

Pricing and staffing should follow this swing.

Early and flexible bookings cancel most





46.9%
Groups cancel

45.7%
Online TA cancel

8.7%
Corporate cancel

61.2%
revenue from India

Four recommendations

Require deposits early

Use deposits or non-refundable rates for bookings made far in advance.

Review overbooking limits

Plan for higher cancellations from group and online agent bookings.

Price for the seasons

Raise rates around the December peak and discount in July.

Grow direct and corporate

These segments cancel the least, so they give steadier revenue.

Limitations and next steps

Limitations

- Simulated data shows typical hotel behaviour, not a real company
- Occupancy uses a fixed room count per hotel
- Revenue is counted in the month of arrival

Next steps

- Cancellation prediction model
- Monthly demand forecast
- Market-segment page and mobile layout

Thank you

Questions welcome

Your Name

github.com/your-username/hotel-booking-bi-dashboard